CONNECTEDNESS

Def: Let X be a top. space. A "separation" of X is a pair U, V of open sets s.t. $U \cap V = \emptyset$ and $U \cup V = X$.

X is said to be "connected" if there exists no separation of X.

Prop: X is connected $\iff$ the only subsets of X that are both open and closed are $\emptyset$ and X.

Lemma: If $Y \subseteq X$, a separation of Y is a pair A, B s.t.
$A \cap B = \emptyset$, $A \cup B = Y$,
$\mathrm{Cl}(A) \cap B = \emptyset$ and
$\mathrm{Cl}(B) \cap A = \emptyset$.

Proof:

Eg: 1. $X = \{0,1\}$, $\mathcal{T} = \{\emptyset, X\}$.

2. For $Y = [-1,0) \cup (0,1]$, the subsets $U = [-1,0)$ and $V = (0,1]$ form a separation.

3. $X = [-1,1]$ is connected (Proof: $\approx$ a lite-section)

4. $\mathbb{Q}$ is not connected. The only connected subsets are finite or singleton sets.
Proof:

5. Any infinite set with cofinite topology is connected.

Lemma: If $C \subseteq D$ form a separation of X and $Y \subseteq X$ is connected, then either $Y \subseteq C$ or $Y \subseteq D$.

Proof:

Theorem: The union of a subfamily of connected subsets of X that all share a point in common is connected.

Proof:

Theorem: Let $A \subseteq X$ be connected.
If B is s.t. $A \subseteq B \subseteq \mathrm{Cl}(A)$, then B is connected.

Proof:

Theorem: Suppose X is connected and $f: X \to Y$ is continuous. Then $f(X)$ is connected.

Proof:

Theorem: A finite product of connected spaces is connected.

Proof:

Non-eg (Infinite product): $\mathbb{R}^{\mathbb{N}}$ with box topology.
But $\mathbb{R}^{\mathbb{N}}$ with product topology is connected.

Proof:

Def: A space is <u>totally disconnected</u> if the only connected subspaces are singletons.
Eg discrete spaces, $\mathbb{Q}$.
<u>Cantor set</u>:

Connected Subsets of $\mathbb{R}$:

Def: A totally ordered set $(X, \leq)$ with more than one element is called a "linear continuum" if the following conditions hold:

1. L has the least upper bound property [every non-empty subset that is bounded above has a least upper bound in L]

2. If $x < y$, $\exists z$ s.t. $x < z < y$.

Theorem: If L is a linear continuum in the order topology [i.e., the topology generated by the basis $\{(a,b) : a, b \in L\}$], then L is connected, and so are intervals (a, b) and rays $\mathcal{T}_a = \{y : x < y\}$ and $\mathcal{T}_b = \{y : y < x\}$.

Proof:

Corollary: $\mathbb{R}$ is connected, and so are intervals and rays in $\mathbb{R}$.

Theorem: [Intermediate Value Theorem]. Let $f: X \to Y$ be a cts map, where X is a connected space and Y is an ordered set in the order topology. If $a, b \in X$ and $r \in Y$ is s.t. $r \in (f(a), f(b))$, then $\exists c \in (a,b)$ s.t. $f(c) = r$.

Proof:

Def: Let $x, y \in X$. A "path" from x to y is a continuous map $f: [a,b] \to X$ for some $[a,b] \subseteq \mathbb{R}$, s.t. $f(a) = x$ and $f(b) = y$.

A space X is said to be "path connected" if every pair of points in X can be joined by a path in X.

Eg

Non-eg